

A Rank-One Splitting Counterexample to Tang–Zhang’s Schatten-Norm Conjecture

Automated mathematics run `fa_banach_001`, lane 9

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Status. Candidate counterexample, likely valid, pending expert review. For every $1 < p < 2$, Conjecture 3.1 of Tang–Zhang is false already for two real 2×2 rank-one matrices.

1 Source problem

Tang and Zhang [1] define $c_p(m)$ to be the least constant such that

$$\left\| \sum_{k=1}^m A_k \right\|_p \leq c_p(m) \left\| \sum_{k=1}^m |A_k| \right\|_p \quad (A_k \in M_n(\mathbb{C})),$$

where $|A| = (A^*A)^{1/2}$ and $\|\cdot\|_p$ is the Schatten p -norm. Their Conjecture 3.1 proposes

$$c_p(m) = \frac{\sqrt{x_{p,m}(x_{p,m} + m - 1)}}{(x_{p,m}^p + m - 1)^{1/p}}, \quad x_{p,m}^p = 2x_{p,m} + m - 1. \quad (1)$$

The conjecture and its rank-one heuristic occur on page 4 of the source PDF.

2 Proof intuition

The source heuristic uses matrices $A_k = uv_k^*$ with a *common* left vector u . Their sum therefore has rank one. That is stable at the Hilbert–Schmidt exponent $p = 2$, but it cannot be extremal below 2.

Indeed, rotate one left vector away from the other by a small amount δ . The principal singular value suffers only a linear loss in δ , while a new singular value is born with squared size proportional to δ . Its contribution to the p -th power of the Schatten norm is therefore proportional to $\delta^{p/2}$. When $1 < p < 2$, we have $p/2 < 1$, so $\delta^{p/2}$ dominates the linear loss. The denominator is unchanged, because it depends only on the right vectors. This turns every proposed two-matrix extremizer into a strict violation.

3 Counterexample theorem

Theorem 1. *For every $1 < p < 2$, the value proposed by (1) is strictly smaller than $c_p(2)$. More precisely, there exist real rank-one matrices $A_1, A_2 \in M_2(\mathbb{R})$ such that*

$$\frac{\|A_1 + A_2\|_p}{\||A_1| + |A_2|\|_p} > \frac{\sqrt{x(x+1)}}{(x^p + 1)^{1/p}}, \quad x^p = 2x + 1.$$

Consequently Conjecture 3.1 of Tang–Zhang is false.

3. CONCLUDING REMARKS

We now turn to the generalized version of Lee's problem, that is, Question [1.3](#). Supported by numerical experiments, we are led to the following conjectural formula for $c_p(m)$.

Conjecture 3.1. *Let $p > 1$ and $m \geq 2$. Let $c_p(m)$ be the smallest number such that*

$$\left\| \sum_{k=1}^m A_k \right\|_p \leq c_p(m) \left\| \sum_{k=1}^m |A_k| \right\|_p \quad \text{for all } A_1, \dots, A_m \in M_n(\mathbb{C}).$$

Then [1](#)

$$c_p(m) = \frac{\sqrt{x_{p,m}(x_{p,m} + m - 1)}}{(x_{p,m}^p + m - 1)^{1/p}}, \quad x_{p,m} > 0 \text{ solves } x^p - 2x - (m - 1) = 0. \quad (3.1)$$

A heuristic derivation of Conjecture [3.1](#). We again employ the construction that realizes equality in the proof of Theorem [1.1](#). Keep the rank-one equiangular family $A_k = u x_k^*$, $\langle x_j, x_k \rangle = s \in (0, 1)$. As $\sum_k A_k = u \sum_k x_k^*$ is rank one, its Schatten p -norm equals its unique singular value:

$$\left\| \sum_{k=1}^m A_k \right\|_p = \left\| \sum_{k=1}^m x_k \right\|_2 = \sqrt{m + m(m-1)s}.$$

The positive operator $\sum_k |A_k| = \sum_k x_k x_k^*$ has eigenvalues $1 + (m-1)s$ (multiplicity 1) and $1 - s$ (multiplicity $m-1$), so

$$\left\| \sum_{k=1}^m |A_k| \right\|_p = \left((1 + (m-1)s)^p + (m-1)(1-s)^p \right)^{1/p}.$$

Maximizing the ratio over $s \in (0, 1)$ is equivalent to introducing $y = \frac{1+(m-1)s}{1-s} \in (1, \infty)$ and optimizing

$$R_p(y) = \frac{\sqrt{y(y+m-1)}}{(y^p + m - 1)^{1/p}}.$$

Figure 1: Conjecture 3.1 and the proposed rank-one extremal family in Tang–Zhang [1, p. 4].

Proof. Fix $1 < p < 2$, put $q = p/2 \in (1/2, 1)$, and let $x > 1$ be the positive solution of $x^p = 2x + 1$. Set

$$\beta = \frac{x-1}{x+1} \in (0, 1).$$

For a parameter $\alpha \in (0, 1]$, take unit vectors in $\mathbb{R}^2$

$$u_1 = v_1 = e_1, \quad u_2 = \alpha e_1 + \sqrt{1 - \alpha^2} e_2, \quad v_2 = \beta e_1 + \sqrt{1 - \beta^2} e_2$$

and define

$$A_1 = u_1 v_1^\top, \quad A_2 = u_2 v_2^\top.$$

Since u_i and v_i are unit vectors,

$$A_i^* A_i = v_i v_i^\top, \quad |A_i| = v_i v_i^\top.$$

The two eigenvalues of $|A_1| + |A_2|$ are $1 + \beta$ and $1 - \beta$.

Write $U = [u_1 \ u_2]$ and $V = [v_1 \ v_2]$. Then $A_1 + A_2 = UV^\top$, and the left and right Gram matrices are

$$G_U = U^\top U = \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix}, \quad G_V = V^\top V = \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix}.$$

The squared singular values of $UV^\top$ are the eigenvalues of $G_U G_V$. These Gram matrices are simultaneously diagonalized by $(1, 1)^\top$ and $(1, -1)^\top$, so the squared singular values are

$$(1 + \alpha)(1 + \beta), \quad (1 - \alpha)(1 - \beta).$$

It follows that the p -th power of the ratio is

$$\mathcal{R}_p(\alpha, \beta)^p = \frac{((1 + \alpha)(1 + \beta))^q + ((1 - \alpha)(1 - \beta))^q}{(1 + \beta)^p + (1 - \beta)^p}. \quad (2)$$

At $\alpha = 1$, the substitution $(1 + \beta)/(1 - \beta) = x$ gives directly

$$\mathcal{R}_p(1, \beta) = \frac{\sqrt{x(x+1)}}{(x^p + 1)^{1/p}},$$

which is exactly the conjectured value for $m = 2$.

We now split the common left vector. Let $\alpha = 1 - \delta$ with $0 < \delta < 1$. The change in the numerator of (2) is

$$D(\delta) = (1 - \beta)^q \delta^q - (1 + \beta)^q (2^q - (2 - \delta)^q).$$

For $0 < q < 1$ and $0 < \delta < 1$,

$$2^q - (2 - \delta)^q = \int_{2-\delta}^2 q t^{q-1} dt \leq q\delta,$$

because $t \geq 1$ throughout the interval. Hence

$$D(\delta) \geq (1 - \beta)^q \delta^q - q(1 + \beta)^q \delta. \quad (3)$$

Define

$$T = \left[\frac{1}{q} \left(\frac{1 - \beta}{1 + \beta} \right)^q \right]^{1/(1-q)}, \quad \delta = \frac{1}{2} \min\{1, T\}.$$

Then $0 < \delta < 1$ and $\delta < T$. The right-hand side of (3) is therefore strictly positive. The denominator in (2) has not changed, so

$$\mathcal{R}_p(1 - \delta, \beta) > \mathcal{R}_p(1, \beta).$$

The matrices constructed above consequently violate the proposed constant. $\square$

4 Numerical illustration and verification

The proof is exact and requires no computation. As a transparent numerical illustration, take $p = 4/3$, $\alpha = 9/10$, and $\beta = 13/20$. Thus

$$A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A_2 = \frac{1}{200} \begin{pmatrix} 117 & 9\sqrt{231} \\ 13\sqrt{19} & \sqrt{4389} \end{pmatrix}.$$

The squared singular values of $A_1 + A_2$ are $627/200$ and $7/200$, while the eigenvalues of $|A_1| + |A_2|$ are $33/20$ and $7/20$. A 100-digit evaluation gives a ratio

$$1.01791549015466577995682703563,$$

whereas the conjectured value is

$$1.01375834364417921376942970795.$$

The deterministic script `code/verify_rank_one_split.py` checks the closed formulas at six representative exponents and this illustration. The seeded exploratory script `code/lee_cp_extremal_search.py` searches arbitrary small complex matrices; it is included only to document how the analytic pattern was found.

5 Scope, novelty, and review recommendation

The theorem disproves the proposed formula for every $1 < p < 2$ already at $m = n = 2$. It does not determine the true constant $c_p(2)$, decide the formula for $p > 2$, or determine $c_p(m)$ for general m .

The local run indexes were searched for the arXiv id, exact title, Lee’s conjecture, and the scalar equation in (1). Bounded web searches on August 9, 2026 used the exact title, the DOI together with “counterexample,” the title together with “counterexample Schatten p ,” and the exact scalar equation. They located the accepted source version and publisher record but no later resolution or this rank-one-splitting argument. Novelty confidence is therefore positive but conservative pending expert literature review.

Human review is strongly recommended. The key points to check are the two Gram-matrix spectral calculations and the scalar estimate $2^q - (2 - \delta)^q \leq q\delta$.

References

- [1] Q. Tang and S. Zhang, *Generalizing Lee’s conjecture on the sum of absolute values of matrices*, Linear Algebra Appl. **731** (2026), 196–204; arXiv:2510.16846v2; doi:10.1016/j.laa.2025.11.015.